Shreya Prakash

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Research Interests: Causal Inference/Discovery

Website: https://shreyap18.github.io/

EDUCATION

University of Washington — Expected Spring 2026

PhD in Statistics, Advanced Data Science Track *(Advisor: Elena Erosheva)*

Thesis: "Statistical Advancements in Causal Decomposition and Discovery for Real-World Applications"

Coursework: Causal Inference, Statistical Learning, Advanced Statistical Theory, Bayesian Statistics, Advanced Regression, Fairness in Machine Learning, Computational Biology, Measure Theory

Carnegie Mellon University — Fall 2019

B.S in Statistics and Machine Learning, University Honors

Coursework: Machine Learning, Statistical Graphics and Visualization, Parallel and Sequential Data Structures and Algorithms (C/SML), Linear Algebra

PROFESSIONAL EXPERIENCE

Applied Science Intern, *Amazon* — Summer 2024, 2025

- Developed novel Bayesian methodology to reduce noise in feature-impact estimates for A/B tests in Weblab, Amazon's internal experimentation platform supporting 100K+ large-scale experiments annually.
- Developed instrumental-variable survival models to study causal factors of workforce attrition as part of Amazon's Execution Planning Science (EPS) team, which develops science solutions for labor planning and operations optimization.

Data Scientist, *Marinus Analytics* — Winter - Summer 2020, Summer 2021

- Applied machine learning and time series analysis for unstructured child welfare case records.
- Launched spam filter and underage person detection algorithms for TraffickJam: an application that uses human trafficking advertisement data to aid law enforcement with finding trafficking victims and traffickers.
- Productionalized Infoshield: a text clustering algorithm for large scale human trafficking advertisement datasets.

Data Science and Research Intern, *84.51°* — Summer 2019

- Fixed issues and tested optimization algorithms for grocery promotion; recommended running promotion optimization for 52 weeks to increase category performance by 4%.

Software Developer Intern, *Optum Technologies* — Summer 2018

- Built authentication, UI and containerized existing application for cloud deployment, generating millions in savings and revenue.

Research Intern, *Royal Caliber D3M Program DARPA* — Summer 2017

- Worked on machine learning on graph datasets and implemented a significantly more efficient way to estimate the number of triangles in a graph, (from $O(V^3)$ to $O(V)$, where V is the number of graph vertices), using a wedge sampling algorithm.

TALKS, PUBLICATIONS & MEDIA

1. **Prakash, S**., Erosheva, E., & Lee, C. (2025). *Causal decomposition of Black-white disparities in NIH's selection of proposals for panel discussion.* (In review). Student Paper Competition Winner (ASA SSS/GSS/SRMS 2026).

2. **Prakash, S**., Xia, F., & Erosheva, E. (2026). *Hypothesis testing for functional causal discovery.* (In review.)
   - Presented at the American Causal Inference Conference (ACIC), 2024, and the Western North American Region of the International Biometric Society (WNAR), 2023.

3. **Prakash, S.**, Xia, F., & Erosheva, E. (2026). *Statistical causal discovery via subsampling.* (In preparation).

4. Samir, F., Ahn, E., **Prakash, S.**, Soskuthy, M., Shwartz, V., & Zhu, J. (2025). *Efficiently identifying low-quality language subsets in multilingual datasets: A case study on a large-scale multilingual audio dataset. Transactions of the Association for Computational Linguistics (TACL).* (Published). https://doi.org/10.1162/tacl_a_00759

5. **Prakash, S.**, Javed, I., Lu, S., Adler, A., LeRoy, B., & Nugent, R. (2019). *Characterizing incidents at Black & Veatch.* Carnegie Mellon University Meeting of the Minds Undergraduate Research Symposium. (Conference presentation).

6. **Prakash, S.**, & Freeman, P. (2019). *Linking galaxies across time via conditional density estimation.* Carnegie Mellon University Meeting of the Minds Undergraduate Research Symposium. (Conference presentation).

7. Konam, S., **Prakash, S.**, Papp, S., Mishra, S., Liu, X., Ma, Z., & Siripurapu, K. (2017). *New app for indigenous farmers. The Hans India.* (Media coverage). https://www.thehansindia.com/

RESEARCH EXPERIENCE

University of Washington

Dissertation Research Winter 2022 - Present

- Developing statistical methods for causal structure learning and causal decomposition.

Research Assistant in WA Notify project Spring 2021 - Fall 2021

- Conducted research on the impact of the privacy-protected exposure notification app (WA Notify) on COVID-19 transmission and identified factors influencing willingness to quarantine and get tested using statistical methods.

Carnegie Mellon University

Undergraduate Research Assistant advised by Alexandra Chouldechova Fall 2019

- Assessed the presence of age, race, or gender-based disparity in the utilization of fully or semi-automated decision-making processes for determining when a case worker should investigate specific abuse cases.

Undergraduate Research Assistant advised by Peter Freeman Spring 2019

- Developed a data pipeline to aid astronomers in understanding the evolution of galaxies based on their current appearances, employing techniques to address imbalanced data.

Undergraduate Research Intern in Black & Veatch Corporate Capstone Project Fall 2018 - Spring 2019

- Created an R Shiny app for analyzing historical company data, predicting injury and property damage cases, and generating prevention strategies through partial dependence plots (pdp).

Undergraduate Research Intern for the KONAM Foundation Fall 2017

- Designed and implemented a machine learning algorithm that assesses the risk of planting certain crops for marginalized farmers in India.

SKILLS

Programming: Proficient in R, Python, SQL. Familiar with C, Standard ML, Matlab, Java, Bash, Mathematica

Libraries/Software: numpy, pandas, scipy, sklearn, statsmodels, tensorflow, seaborn, matplotlib, rjags, tidyverse, parallel, Git, Docker, AWS S3 & EC2, Keras

AWARDS

- Student Paper Competition Winner, American Statistical Association Sections on Social Statistics, Government Statistics, and Survey Research Methods 2026
- Rising Star Award, Oden Institute for Computational Engineering and Sciences 2025
- Travel Award, Center for Statistics and the Social Sciences (University of Washington) 2023

- Third Place, Poster Presentation Competition, Meeting of Minds Undergraduate Research Symposium 2019

TEACHING EXPERIENCE

Teaching Assistant, *University of Washington (2021 - Present), Carnegie Mellon University (2017 - 2019)*

- Introduction to Mathematical Statistics (STAT 509), Applied Regression (STAT 504), Quantitative Introductory Statistics for Data Science (STAT 391), Elements of Statistical Methods (STAT 311), Statistical Reasoning (STAT 220), Causal Modeling (STAT 566), Statistical Concepts and Methods for the Social Sciences (STAT 221), Introduction to Probability Theory (36-225), Introduction to Machine Learning (10-601), Methods for Statistics and Data Science (36-202)